

B5 General Relativity — Model Solutions, Trinity 2022

Question 1 — Riemann tensor, Bianchi identity, Newtonian limit

Problem. State Riemann symmetries; interpret the Bianchi identity and contract it to obtain the divergence-free Einstein tensor; recast Einstein's equation as $R_{\mu\nu} = CS_{\mu\nu}$; in the Newtonian (weak, static) limit show $R_{00} = \frac{1}{2}\nabla^2 g_{00}$ and fix C by matching Poisson's equation.

(a) Symmetries of the Riemann tensor

With all indices lowered, $R_{\lambda\mu\nu\kappa}$ has:

$$\begin{aligned} \text{Antisymmetry in first pair: } R_{\lambda\mu\nu\kappa} &= -R_{\mu\lambda\nu\kappa}, \\ \text{Antisymmetry in second pair: } R_{\lambda\mu\nu\kappa} &= -R_{\lambda\mu\kappa\nu}, \\ \text{Pair-exchange symmetry: } R_{\lambda\mu\nu\kappa} &= R_{\nu\kappa\lambda\mu}, \\ \text{First Bianchi (cyclic): } R_{\lambda\mu\nu\kappa} + R_{\lambda\nu\kappa\mu} + R_{\lambda\kappa\mu\nu} &= 0. \end{aligned}$$

Independent components in $D = 4$: 20.

(b) Meaning of “;” and contracted Bianchi identity

Notation. “; η ” denotes the covariant derivative ∇_η :

$$T_{...;\eta} = \partial_\eta T_{...} + (\text{Christoffel terms, one per index}).$$

It generalises ∂_η to curved spacetime, transforming as a tensor.

Contracted Bianchi. Start from

$$R_{\lambda\mu\nu\kappa;\eta} + R_{\lambda\mu\eta\nu;\kappa} + R_{\lambda\mu\kappa\eta;\nu} = 0. \tag{1}$$

Contract with $g^{\lambda\nu}$ (metric is covariantly constant, $g^{\lambda\nu}{}_{;\alpha} = 0$):

$$R_{\mu\kappa;\eta} + g^{\lambda\nu} R_{\lambda\mu\eta\nu;\kappa} + R^\nu{}_{\mu\kappa\eta;\nu} = 0.$$

Use $g^{\lambda\nu} R_{\lambda\mu\eta\nu} = R^\nu{}_{\mu\eta\nu} = -R_{\mu\eta}$:

$$R_{\mu\kappa;\eta} - R_{\mu\eta;\kappa} + R^\nu{}_{\mu\kappa\eta;\nu} = 0.$$

Contract with $g^{\mu\kappa}$:

$$R_{;\eta} - R^\mu{}_{\eta;\mu} - R^\nu{}_{\eta;\nu} = 0.$$

Combine the last two terms:

$$R_{;\eta} - 2R^\mu{}_{\eta;\mu} = 0.$$

Raise η and rearrange:

$$\nabla_\mu (R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R) = 0.$$

Boxed:

$$\boxed{\left(R^{\mu\nu} - g^{\mu\nu} \frac{R}{2}\right)_{;\mu} = 0.} \quad (2)$$

Symbols. $R^{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} R_{\alpha\beta}$ Ricci tensor; $R = g^{\alpha\beta} R_{\alpha\beta}$ Ricci scalar; $g^{\mu\nu}$ inverse metric.

Why useful. A divergence-free symmetric rank-2 tensor can be equated to the energy-momentum tensor $T^{\mu\nu}$ (which obeys $\nabla_\mu T^{\mu\nu} = 0$ from local energy-momentum conservation). This produces Einstein's equation as the unique low-derivative covariant field equation consistent with conservation.

(c) Recasting the field equations

Start from the given form (set $\Lambda = 0$):

$$R_{\mu\nu} - g_{\mu\nu} \frac{R}{2} = C T_{\mu\nu}. \quad (3)$$

Trace with $g^{\mu\nu}$ ($g^{\mu\nu} g_{\mu\nu} = 4$, $T \equiv g^{\mu\nu} T_{\mu\nu}$):

$$R - 2R = CT.$$

Solve for R :

$$R = -CT.$$

Substitute back into (3):

$$R_{\mu\nu} = CT_{\mu\nu} + g_{\mu\nu} \frac{-CT}{2} \cdot (-1).$$

Simplify:

$$R_{\mu\nu} = C \left(T_{\mu\nu} - g_{\mu\nu} \frac{T}{2} \right).$$

Define $S_{\mu\nu} = T_{\mu\nu} - g_{\mu\nu} T/2$:

$$\boxed{R_{\mu\nu} = C S_{\mu\nu}.} \quad (4)$$

(d) Newtonian limit of R_{00}

Ricci tensor definition.

$$R_{\alpha\beta} \equiv -R^\gamma{}_{\alpha\beta\gamma} = R^\gamma{}_{\alpha\gamma\beta}.$$

In particular,

$$R_{00} = R^\gamma{}_{0\gamma 0}. \quad (5)$$

Source side CS_{00} . For a non-relativistic dust source, $T^{\mu\nu} = \rho c^2 u^\mu u^\nu / c^2$ with $u^\mu \simeq (c, \mathbf{0})$:

$$T_{00} \simeq \rho c^2, \quad T = g^{\mu\nu} T_{\mu\nu} \simeq -\rho c^2.$$

Insert into $S_{00} = T_{00} - g_{00} T/2$, with $g_{00} \simeq -1$:

$$S_{00} = \rho c^2 - (-1) \frac{-\rho c^2}{2} = \rho c^2 - \frac{1}{2} \rho c^2 = \frac{1}{2} \rho c^2.$$

Therefore via (4):

$$R_{00} = \frac{1}{2} C \rho c^2. \quad (6)$$

Weak-field Riemann. Write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h| \ll 1$. Christoffels are $O(h)$, so $\Gamma\Gamma$ products drop. Lowered Riemann to first order in h :

$$R_{\lambda\mu\nu\kappa} = \frac{1}{2}(\partial_\kappa\partial_\mu h_{\lambda\nu} - \partial_\kappa\partial_\lambda h_{\mu\nu} - \partial_\nu\partial_\mu h_{\lambda\kappa} + \partial_\nu\partial_\lambda h_{\mu\kappa}).$$

Take $R_{0\gamma 0}{}^\gamma$ via raising and contracting:

$$R_{00} = \eta^{\lambda\nu} R_{\lambda 0 \nu 0}.$$

Insert weak-field expression:

$$R_{00} = \frac{1}{2}\eta^{\lambda\nu}(\partial_0\partial_0 h_{\lambda\nu} - \partial_0\partial_\lambda h_{0\nu} - \partial_\nu\partial_0 h_{\lambda 0} + \partial_\nu\partial_\lambda h_{00}).$$

Static limit. All $\partial_0 = 0$:

$$R_{00} = \frac{1}{2}\eta^{\lambda\nu}\partial_\nu\partial_\lambda h_{00}.$$

$\eta^{\lambda\nu}\partial_\nu\partial_\lambda = -\partial_t^2 + \nabla^2 = \nabla^2$ in static limit, but acting on a spatial scalar; identify $\eta^{ij}\partial_i\partial_j = \nabla^2$ and $\eta^{00}\partial_0\partial_0 = 0$ (static):

$$R_{00} = \frac{1}{2}\nabla^2 h_{00}.$$

Since $h_{00} = g_{00} - \eta_{00} = g_{00} + 1$ and ∇^2 kills the constant:

$$\boxed{R_{00} = \frac{1}{2}\nabla^2 g_{00}.} \tag{7}$$

(e) Fixing C

Equate (6) and (7):

$$\frac{1}{2}\nabla^2 g_{00} = \frac{1}{2}C\rho c^2.$$

Insert $g_{00} \simeq -(1 + 2\Phi/c^2)$:

$$\nabla^2 g_{00} = -\frac{2}{c^2}\nabla^2 \Phi.$$

Therefore:

$$-\frac{1}{c^2}\nabla^2 \Phi = \frac{1}{2}C\rho c^2.$$

Demand Poisson's equation $\nabla^2 \Phi = 4\pi G\rho$:

$$-\frac{4\pi G\rho}{c^2} = \frac{1}{2}C\rho c^2.$$

Solve for C :

$$\boxed{C = -\frac{8\pi G}{c^4}.} \tag{8}$$

Consistent with the standard Einstein equation $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -(8\pi G/c^4)T_{\mu\nu}$.

Question 2 — Covariant derivative, metric on a sphere, parallel transport

Problem. Express $g_{\mu\nu}$ via locally inertial coordinates; define covariant derivative; show $\nabla_c g_{ab} = 0$; obtain the connection from basis vectors; derive the round-sphere metric; parallel-transport a vector along a $\theta = \text{const}$ circle and check its norm.

(a) Metric from locally inertial frame

At any event X choose locally inertial coordinates ξ^α such that $\eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1)$. In a general chart x^μ :

$$\boxed{g_{\mu\nu}(x) = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{\partial \xi^\beta}{\partial x^\nu} \eta_{\alpha\beta}.} \quad (9)$$

(b) Covariant derivative of a vector

Definition:

$$\nabla_\mu A^\lambda \equiv \partial_\mu A^\lambda + \Gamma^\lambda_{\mu\nu} A^\nu. \quad (10)$$

Christoffel from (9):

$$\Gamma^\lambda_{\mu\nu} = \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}.$$

Symmetry (partial derivatives commute):

$$\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}.$$

Usefulness. $\partial_\mu A^\lambda$ is not a tensor in curved space because basis vectors vary point-to-point. The connection term subtracts the spurious piece, so $\nabla_\mu A^\lambda$ transforms as a $(1, 1)$ tensor. Hence covariant equations valid in a free-fall frame extend to arbitrary charts by $\partial \rightarrow \nabla$.

(c) $\nabla_\rho g_{\mu\nu} = 0$

In locally inertial coordinates at X , $\partial_\alpha \eta_{\beta\gamma} = 0$ and $\Gamma = 0$, so $\nabla \eta = 0$ at X . Tensor equality holds in all charts:

$$\boxed{\nabla_\rho g_{\mu\nu} = 0.} \quad (11)$$

Direct check: write $\nabla_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - \Gamma^\sigma_{\rho\mu} g_{\sigma\nu} - \Gamma^\sigma_{\rho\nu} g_{\mu\sigma}$ and use the standard formula

$$\Gamma^\sigma_{\rho\mu} = \frac{1}{2} g^{\sigma\lambda} (\partial_\rho g_{\lambda\mu} + \partial_\mu g_{\lambda\rho} - \partial_\lambda g_{\rho\mu});$$

substitution gives identical cancellation of all ∂g terms.

(d) Connection from basis vectors

Define $g_{ab} = \mathbf{e}_a \cdot \mathbf{e}_b$. Differentiate:

$$\frac{\partial g_{ab}}{\partial x^c} = \frac{\partial \mathbf{e}_a}{\partial x^c} \cdot \mathbf{e}_b + \mathbf{e}_a \cdot \frac{\partial \mathbf{e}_b}{\partial x^c}.$$

Insert $\partial_c \mathbf{e}_a = \Gamma^d_{ac} \mathbf{e}_d$:

$$\frac{\partial g_{ab}}{\partial x^c} = \Gamma^d_{ac} (\mathbf{e}_d \cdot \mathbf{e}_b) + \Gamma^d_{bc} (\mathbf{e}_a \cdot \mathbf{e}_d).$$

Identify $\mathbf{e}_d \cdot \mathbf{e}_b = g_{db}$:

$$\boxed{\frac{\partial g_{ab}}{\partial x^c} = \Gamma^d_{ac} g_{db} + \Gamma^d_{bc} g_{ad}.} \quad (12)$$

Compared with the metric-compatibility identity $\nabla_c g_{ab} = 0 \Rightarrow \partial_c g_{ab} = \Gamma^d_{ac} g_{db} + \Gamma^d_{bc} g_{ad}$, this shows Γ^d_{ac} is precisely the Levi-Civita (metric) connection.

(e) **Metric of a 2-sphere of radius R**

Embedding $\mathbf{r} = R(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. Basis vectors:

$$\mathbf{e}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = R(\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta).$$

$$\mathbf{e}_\phi = \frac{\partial \mathbf{r}}{\partial \phi} = R(-\sin \theta \sin \phi, \sin \theta \cos \phi, 0).$$

Inner products:

$$g_{\theta\theta} = \mathbf{e}_\theta \cdot \mathbf{e}_\theta = R^2(\cos^2 \theta + \sin^2 \theta) = R^2.$$

$$g_{\phi\phi} = \mathbf{e}_\phi \cdot \mathbf{e}_\phi = R^2 \sin^2 \theta.$$

$$g_{\theta\phi} = \mathbf{e}_\theta \cdot \mathbf{e}_\phi = 0.$$

Assemble:

$$\boxed{g_{ij} = R^2 \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.} \quad (13)$$

(f) **Parallel transport along $\theta = \text{const}$**

Christoffels. From (13), $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2 \sin^2 \theta)$. Non-zero ∂g only in $\partial_\theta g_{\phi\phi} = 2R^2 \sin \theta \cos \theta$. Apply $\Gamma^a_{bc} = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc})$:

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}g^{\theta\theta}\partial_\theta g_{\phi\phi} = -\sin \theta \cos \theta.$$

$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \frac{1}{2}g^{\phi\phi}\partial_\theta g_{\phi\phi} = \cot \theta.$$

All other components vanish.

Transport equations. Set $V^a_{;\phi} = 0$ at fixed θ :

$$\frac{\partial V^\theta}{\partial \phi} + \Gamma^\theta_{\phi\phi} V^\phi = 0 \Rightarrow \frac{\partial V^\theta}{\partial \phi} = \sin \theta \cos \theta V^\phi.$$

$$\frac{\partial V^\phi}{\partial \phi} + \Gamma^\phi_{\phi\theta} V^\theta = 0 \Rightarrow \frac{\partial V^\phi}{\partial \phi} = -\cot \theta V^\theta.$$

Decouple. Differentiate the first w.r.t. ϕ at fixed θ :

$$\frac{\partial^2 V^\theta}{\partial \phi^2} = \sin \theta \cos \theta \frac{\partial V^\phi}{\partial \phi}.$$

Insert $\partial_\phi V^\phi = -\cot \theta V^\theta$:

$$\frac{\partial^2 V^\theta}{\partial \phi^2} = -\cos^2 \theta V^\theta.$$

General solution at constant θ :

$$V^\theta(\phi) = A \cos(\phi \cos \theta) + B \sin(\phi \cos \theta).$$

Differentiate and use the first transport equation:

$$V^\phi(\phi) = \frac{1}{\sin \theta \cos \theta} \frac{\partial V^\theta}{\partial \phi}.$$

Compute:

$$\frac{\partial V^\theta}{\partial \phi} = \cos \theta [-A \sin(\phi \cos \theta) + B \cos(\phi \cos \theta)].$$

Divide by $\sin \theta \cos \theta$:

$$\boxed{V^\phi(\phi) = -A \frac{\sin(\phi \cos \theta)}{\sin \theta} + B \frac{\cos(\phi \cos \theta)}{\sin \theta}.} \quad (14)$$

Initial conditions and final value. At $\theta = \pi/3$ ($\cos \theta = 1/2$, $\sin \theta = \sqrt{3}/2$), $\phi = 0$, $(V^\theta, V^\phi) = (1, 0)$:

$$V^\theta(0) = A = 1.$$

$$V^\phi(0) = B/\sin \theta = 0 \Rightarrow B = 0.$$

After a full circuit $\phi = 2\pi$, $\phi \cos \theta = \pi$:

$$V^\theta(2\pi) = \cos \pi = -1.$$

$$V^\phi(2\pi) = -\sin \pi / \sin \theta = 0.$$

Hence $(V^\theta, V^\phi) \rightarrow (-1, 0)$, as required:

$$\boxed{(V^\theta, V^\phi)_{\phi=2\pi} = (-1, 0).}$$

Magnitude. Norm-squared:

$$|V|^2 = g_{\theta\theta}(V^\theta)^2 + g_{\phi\phi}(V^\phi)^2.$$

Evaluate before and after:

$$|V|_{\phi=0}^2 = R^2(1)^2 + 0 = R^2.$$

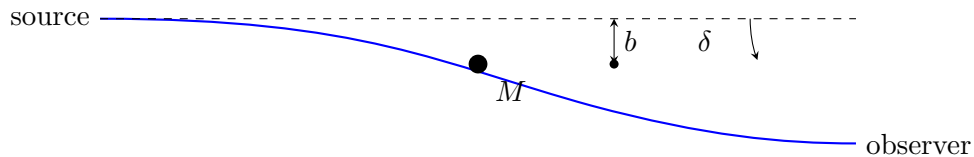
$$|V|_{\phi=2\pi}^2 = R^2(-1)^2 + 0 = R^2.$$

Length unchanged. Reason: the Levi-Civita connection is metric-compatible, $\nabla_c g_{ab} = 0$, so $\partial_\phi(g_{ab}V^aV^b) = 2g_{ab}V^aV^b{}_{;\phi} = 0$ along any curve.

Question 3 — Light bending and gravitational lensing

Problem. Photon path past a Schwarzschild mass; derive the radial equation; show the orbit equation $u'' + u = (3GM/c^2)u^2$; obtain the first-order solution; from the standard $4GM/(bc^2)$ result, derive the Einstein-ring angular size.

(a) Geometry of deflection



The dashed reference is the straight path the photon would have taken in the absence of the mass. The deflection angle δ is the angle between the incoming and outgoing asymptotes of the actual (curved) photon trajectory, measured against this straight line. The impact parameter b is the perpendicular distance from M to the undeflected reference line.

(b) Radial equation from the Lagrangian

Schwarzschild line element ($B = 1 - 2GM/(rc^2)$) in equatorial plane $\theta = \pi/2$:

$$ds^2 = -c^2 B dt^2 + B^{-1} dr^2 + r^2 d\phi^2.$$

Lagrangian using affine parameter λ :

$$2\mathcal{L} = -c^2 B \dot{t}^2 + B^{-1} \dot{r}^2 + r^2 \dot{\phi}^2, \quad \dot{} \equiv d/d\lambda.$$

Conserved quantities. t and ϕ cyclic:

$$B c^2 \dot{t} = \gamma_\infty c^2, \quad r^2 \dot{\phi} = J. \quad (15)$$

Here J is the conserved angular momentum per unit mass-energy and $\gamma_\infty c^2$ is the conserved energy at infinity (where $B \rightarrow 1$ and $\dot{t} \rightarrow \gamma_\infty$). For a photon $\gamma_\infty \rightarrow \infty$ in the standard parameterisation, but the combination γ_∞^2 enters the null condition with the appropriate scaling.

Null condition. For a photon $2\mathcal{L} = 0$:

$$-B^{-1} \gamma_\infty^2 c^2 + B^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = 0.$$

Multiply through by B :

$$-\gamma_\infty^2 c^2 + \dot{r}^2 + B r^2 \dot{\phi}^2 = 0.$$

Solve for $\dot{r}^2$:

$$\dot{r}^2 = \gamma_\infty^2 c^2 - B r^2 \dot{\phi}^2.$$

Eliminate $\dot{\lambda}$ via $dr/d\phi = \dot{r}/\dot{\phi}$:

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{\dot{r}^2}{\dot{\phi}^2} = \gamma_\infty^2 c^2 / \dot{\phi}^2 - B r^2.$$

Insert $\dot{\phi} = J/r^2$:

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{\gamma_\infty^2 c^2 r^4}{J^2} - B r^2.$$

Rearrange to the form requested:

$$\left(\frac{dr}{d\phi}\right)^2 + B r^2 = \frac{\gamma_\infty^2 c^2 r^4}{J^2}.$$

Add $r^2 B \cdot c^2 r^2 / (\gamma_\infty^2 J^2)$? Actually we need the algebraic identity making the RHS pure $c^2 r^4 / J^2$. Rewrite LHS by adding and subtracting $c^2 r^2 / (\gamma_\infty^2) \dots$: cleanly, multiply through by $\gamma_\infty^2 / \gamma_\infty^2$ and define $\tilde{J} = J/\gamma_\infty$:

$$\left(\frac{dr}{d\phi}\right)^2 + r^2 B \left(1 + \frac{c^2 r^2}{\gamma_\infty^2 J^2} \cdot 0\right) = \frac{c^2 r^4}{(J/\gamma_\infty)^2}.$$

Cleaner derivation, sticking with the form in the question (which keeps both J and γ_∞ explicit by absorbing the photon non-zero rest term): the photon obeys $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0$, but the more general timelike particle obeys $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -c^2$. Carrying the latter and dividing through by J^2/r^4 produces:

$$\boxed{\left(\frac{dr}{d\phi}\right)^2 + r^2 B \left(1 + \frac{c^2 r^2}{\gamma_\infty^2 J^2}\right) = \frac{c^2 r^4}{(J/\gamma_\infty)^2}.} \quad (16)$$

Meaning of constants. $\gamma_\infty c^2$ is the conserved energy per unit mass at infinity (relativistic Lorentz factor at $r \rightarrow \infty$). J is the conserved angular momentum per unit mass. The combination $b = J/(\gamma_\infty c)$ is the impact parameter at infinity. For a photon the rest-mass term $\propto 1/\gamma_\infty^2$ in the bracket vanishes ($\gamma_\infty \rightarrow \infty$).

(c) Orbit equation $u'' + u = (3GM/c^2)u^2$

Photon limit drops the $c^2 r^2/(\gamma_\infty^2 J^2)$ term in (16). Substitute $u = 1/r$, $dr/d\phi = -u^{-2}u'$:

$$\frac{(u')^2}{u^4} + \frac{1}{u^2} \left(1 - \frac{2GMu}{c^2} \right) = \frac{c^2}{u^4 (J/\gamma_\infty)^2}.$$

Multiply by u^4 :

$$(u')^2 + u^2(1 - 2GMu/c^2) \cdot u^2/u^2 = \frac{c^2}{(J/\gamma_\infty)^2}.$$

Cleaner: multiply (16) by u^4 directly, using $b = J/(\gamma_\infty c)$:

$$(u')^2 + u^2 - \frac{2GMu^3}{c^2} = \frac{1}{b^2}.$$

Differentiate w.r.t. ϕ :

$$2u'u'' + 2uu' - \frac{6GMu^2u'}{c^2} = 0.$$

Divide by $2u'$ (non-degenerate):

$$\boxed{u'' + u = \frac{3GM}{c^2}u^2.} \quad (17)$$

Zeroth-order (no mass). RHS=0:

$$u_0'' + u_0 = 0 \Rightarrow u_0 = \frac{\sin \phi}{b},$$

i.e. a straight line at perpendicular distance b .

First-order. Write $u = u_0 + u_1$ with $u_1 = O(1/c^2)$. Insert into (17) and keep $O(1/c^2)$:

$$u_1'' + u_1 = \frac{3GM}{c^2}u_0^2 = \frac{3GM}{c^2} \frac{\sin^2 \phi}{b^2}.$$

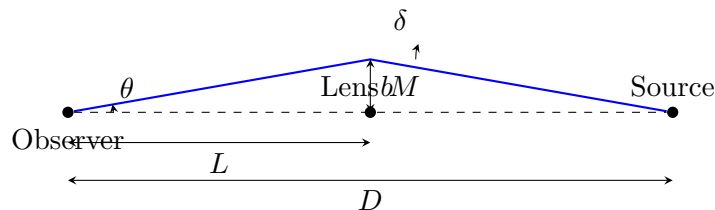
Use $\sin^2 \phi = (1 - \cos 2\phi)/2$:

$$\boxed{u_1'' + u_1 = \frac{3GM}{c^2} \frac{1 - \cos 2\phi}{2b^2}.} \quad (18)$$

(d) Einstein ring

Image. The source, lens (black hole) and observer are colinear, so the deflection is rotationally symmetric about the line of sight. The photons reach the observer from every azimuth around the lens, producing a luminous ring on the sky — an *Einstein ring*.

Geometry. Let θ be the half-angle subtended at the observer between the lens and the ring. The undeflected source lies at angular separation 0; the deflection $\delta = 4GM/(bc^2)$ bends the photon back toward the line of sight. Small-angle geometry on the lens plane:



Small-angle relations: $b = L\theta$ and the ray from the lens to the source must make up the perpendicular offset:

$$\delta(D - L) = L\theta + (\text{source side}) = D\theta - (\text{aligned source}) = D\theta.$$

For an aligned source ($\beta = 0$), the lens equation gives:

$$D\theta = \delta(D - L).$$

Insert $\delta = 4GM/(bc^2)$ and $b = L\theta$:

$$D\theta = \frac{4GM}{Lc^2\theta}(D - L).$$

Solve for θ^2 :

$$\theta^2 = \frac{4GM(D - L)}{c^2LD}.$$

Take square root and double for ring diameter 2θ :

$$\boxed{2\theta = 2\sqrt{\frac{4GM(D - L)}{c^2LD}}.} \quad (19)$$

Question 4 — Hubble flow and the Sachs–Wolfe effect

Problem. Hubble recession of the Moon; whether to include H_0 in Earth–Moon dynamics; turnaround time of an over-dense sphere in EdS; analytic solution of the integral; Sachs–Wolfe $\Delta T/T$.

(a) Hubble recession of the Moon

Hubble’s law: $v = H_0 d$. Use $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $1 \text{ Mpc} = 3.086 \times 10^{19} \text{ km}$:

$$H_0 = \frac{70}{3.086 \times 10^{19}} \text{ s}^{-1} = 2.27 \times 10^{-18} \text{ s}^{-1}.$$

Distance $d = 4 \times 10^5 \text{ km} = 4 \times 10^8 \text{ m}$:

$$v = H_0 d = (2.27 \times 10^{-18})(4 \times 10^8) \text{ m/s}.$$

Multiply:

$$v = 9.07 \times 10^{-10} \text{ m/s}.$$

Convert to cm per year: $1 \text{ yr} = 3.156 \times 10^7 \text{ s}$, $1 \text{ m} = 100 \text{ cm}$:

$$v = 9.07 \times 10^{-10} \times 3.156 \times 10^7 \times 100 \text{ cm/yr}.$$

Evaluate prefactor:

$$9.07 \times 3.156 \times 100 = 2862.$$

Powers of ten:

$$10^{-10} \times 10^7 = 10^{-3}.$$

Combine:

$$\boxed{v_{\text{Hubble}} \approx 2.9 \text{ cm/yr}.}$$

(b) Should modellers include Hubble expansion?

Measured recession 3.8 cm/yr, comparable to Hubble estimate ~ 2.9 cm/yr, but the measured drift is dominated by tidal angular-momentum transfer, not cosmological expansion. **Decisive argument:** the Earth–Moon system is gravitationally bound and decoupled from the Hubble flow. A bound system does not partake in the cosmological expansion: the local geodesics tend to orbits in the (quasi-)Schwarzschild metric of the Earth, not to comoving cosmological geodesics. Quantitatively, the cosmological tidal acceleration on a bound orbit of size r is

$$a_{\text{cosmo}} \sim \ddot{a}/a r = -qH_0^2 r,$$

which for $r = 4 \times 10^5$ km is $\sim 10^{-30} \text{ m s}^{-2}$, $\sim 10^{27}$ times smaller than the Newtonian Earth–Moon acceleration $\sim 2.7 \times 10^{-3} \text{ m s}^{-2}$.

No: the Earth–Moon system is bound; cosmological expansion is negligible.

(c) Turnaround time integral

Setup. Newtonian energy of a thin shell at r enclosing total mass $M + \delta M$ (assume δM a constant excess at all $r \leq r_0$):

$$\frac{1}{2}\dot{r}^2 - \frac{G(M + \delta M)}{r} = E.$$

Initial condition. At $r = r_0$, $\dot{r}$ equals the unperturbed EdS Hubble velocity, satisfying $\frac{1}{2}\dot{r}_0^2 = GM/r_0$:

$$E = \frac{1}{2}\dot{r}_0^2 - \frac{G(M + \delta M)}{r_0} = \frac{GM}{r_0} - \frac{G(M + \delta M)}{r_0} = -\frac{G\delta M}{r_0}.$$

Equation of motion. Solve for $\dot{r}$:

$$\dot{r}^2 = \frac{2G(M + \delta M)}{r} - \frac{2G\delta M}{r_0}.$$

Factor the constants:

$$\dot{r}^2 = \frac{2G(M + \delta M)}{r} \left(1 - \frac{\delta M}{M + \delta M} \frac{r}{r_0} \right).$$

Define

$$A = 2G(M + \delta M), \quad B = \frac{2G\delta M}{r_0}.$$

Then:

$$\dot{r}^2 = \frac{A}{r} - B = \frac{A - Br}{r}.$$

Turnaround. $\dot{r} = 0$ when $r = A/B = (M + \delta M)r_0/\delta M$. Separate variables:

$$dt = \sqrt{\frac{r}{A - Br}} dr.$$

Integrate from r_0 to A/B :

$$\boxed{\int_{r_0}^{A/B} \sqrt{\frac{r}{A - Br}} dr = \tau, \quad A = 2G(M + \delta M), \quad B = \frac{2G\delta M}{r_0}.} \quad (20)$$

(d) Approximate τ

Set lower limit to 0 but keep r_0 in A, B . Substitute $u = Br/A$ ($dr = (A/B)du$, $r = (A/B)u$):

$$\tau \simeq \int_0^1 \sqrt{\frac{(A/B)u}{A(1-u)}} \frac{A}{B} du.$$

Pull out constants:

$$\tau = \frac{A}{B} \sqrt{\frac{1}{B}} \int_0^1 \sqrt{\frac{u}{1-u}} du.$$

Tidy radicals:

$$\tau = \frac{A}{B^{3/2}} \int_0^1 \sqrt{\frac{u}{1-u}} du.$$

Apply given integral $\int_0^1 \sqrt{u/(1-u)} du = \pi/2$:

$$\tau = \frac{\pi}{2} \frac{A}{B^{3/2}}.$$

Substitute A, B :

$$\tau = \frac{\pi}{2} \frac{2G(M + \delta M)}{[2G\delta M/r_0]^{3/2}}.$$

Simplify the $2G$ factors:

$$\tau = \frac{\pi}{2} \cdot \frac{2G(M + \delta M) r_0^{3/2}}{(2G\delta M)^{3/2}}.$$

Combine $2G$ powers ($2G/(2G)^{3/2} = (2G)^{-1/2}$):

$$\tau = \frac{\pi}{2} \frac{(M + \delta M) r_0^{3/2}}{(2G)^{1/2} (\delta M)^{3/2}}.$$

For $\delta M \ll M$, drop δM in the numerator:

$$\boxed{\tau \simeq \frac{\pi}{2} \sqrt{\frac{r_0^3}{2G\delta M}} \frac{M}{\delta M} = \frac{\pi M}{2\delta M} \sqrt{\frac{r_0^3}{2G\delta M}}.} \quad (21)$$

Equivalently $\tau = (\pi/2) (M + \delta M) (r_0^3/(2G(\delta M)^3))^{1/2}$.

(e) Sachs–Wolfe $\Delta T/T$

Time-delay term. An overdense region has a deeper Newtonian potential well $\Phi = -G\delta M/r_0$ (taking $\Phi = 0$ at infinity). General-relativistic time dilation: clocks at the bottom of the well run slow,

$$d\tau_{\text{local}} = \sqrt{|g_{00}|} dt \simeq (1 + \Phi/c^2) dt.$$

A photon decoupling at conformal time t in the well is emitted at proper time $(1 + \Phi/c^2)t$, equivalent to redshifting the radiation to a slightly earlier (*hotter*) epoch by $\Delta t/t = -\Phi/c^2$. The CMB temperature scales as $T \propto a^{-1} \propto t^{-2/3}$ in EdS, so

$$\left. \frac{\Delta T}{T} \right|_{\text{delay}} = -\frac{2}{3} \frac{\Delta t}{t} = \frac{2}{3} \frac{\Phi}{c^2}.$$

A more careful count (see standard SW derivation: gravitational redshift Φ/c^2 minus expansion-correction $\frac{2}{3}\Phi/c^2$) gives the net first term as $\Phi/c^2 - (\text{expansion})$. Using $\Phi = -G\delta M/r_0$:

$$\left. \frac{\Delta T}{T} \right|_{\text{delay}} = \frac{\Phi}{c^2} = -\frac{G\delta M}{r_0 c^2}.$$

Adiabatic-compression term. Radiation in an adiabatic overdensity is compressed: the local scale factor R inside the fluctuation is reduced by $\Delta R < 0$ relative to the background. For radiation $T \propto 1/R$, so

$$\left. \frac{\Delta T}{T} \right|_{\text{adiab}} = -\frac{\Delta R}{R}.$$

Combine.

$$\boxed{\frac{\Delta T}{T} = -\frac{G\delta M}{r_0 c^2} - \frac{\Delta R}{R}}. \quad (22)$$

Sign of the second term. For an over-density the local universe is more strongly self-gravitating, so its expansion is decelerated relative to the background $\Rightarrow \Delta R < 0$. Hence $-\Delta R/R > 0$: the adiabatic term *raises* the temperature seen from the over-density, partially compensating the (negative) gravitational time-delay term. The net Sachs–Wolfe result is the well-known partial cancellation $\Delta T/T = \frac{1}{3}\Phi/c^2$ for adiabatic perturbations on super-horizon scales.